

Advanced Computer Networks – PCS223

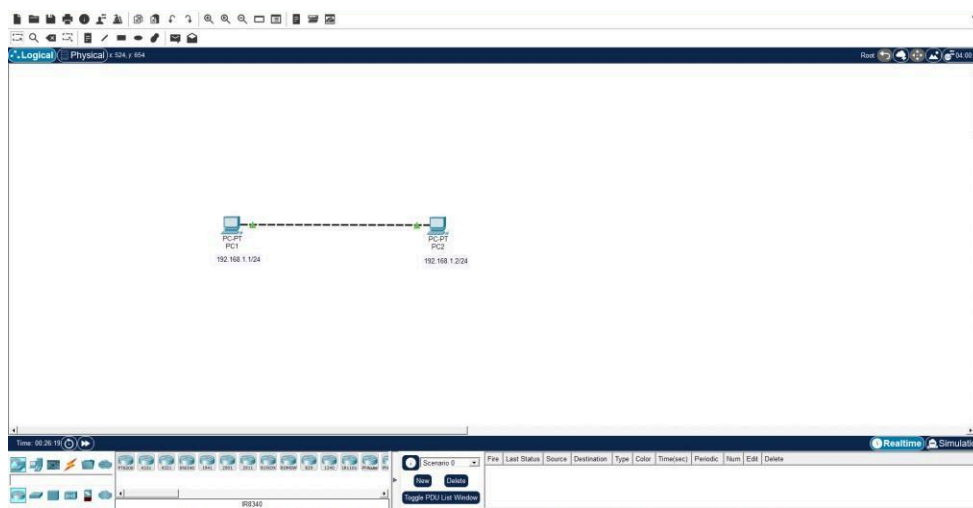
Assignment 3

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Subgroup – CS4

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1. Connect one PC with another PC as per diagram below.



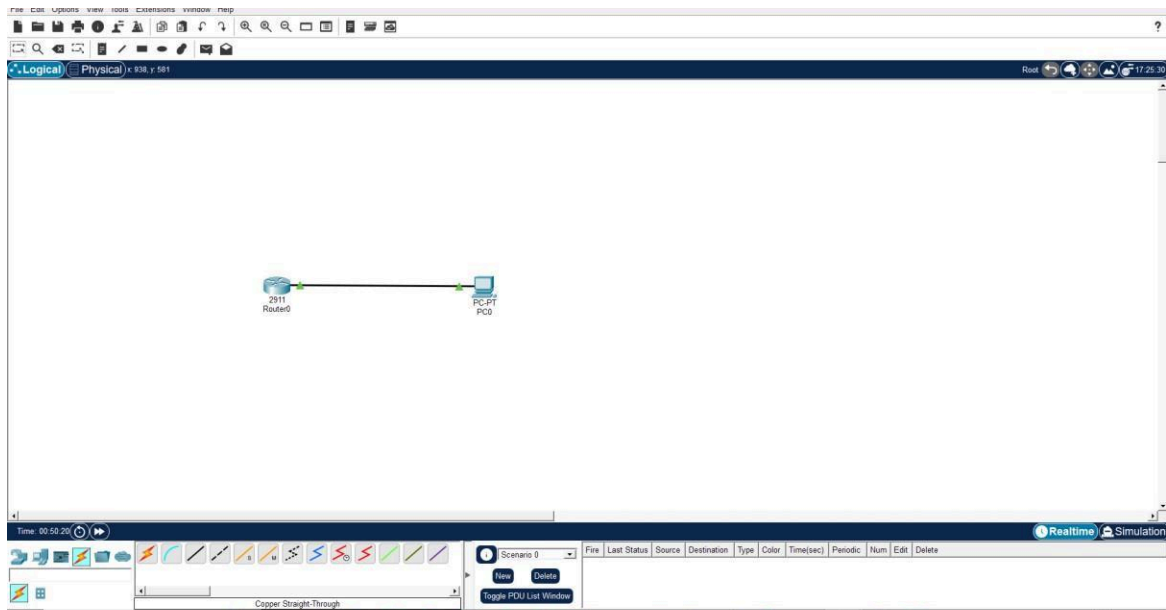
```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

2. Connect a PC with Router Ethernet port as shown below.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:FFFF:FEC7:C1D4
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.128.20.2
    Subnet Mask . . . . .: 255.0.0.0
    Default Gateway . . . . .: ::
                                10.128.20.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

Dialer1 Connection:
--More--
    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:FFFF:FEC7:C1D4
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 10.128.20.1

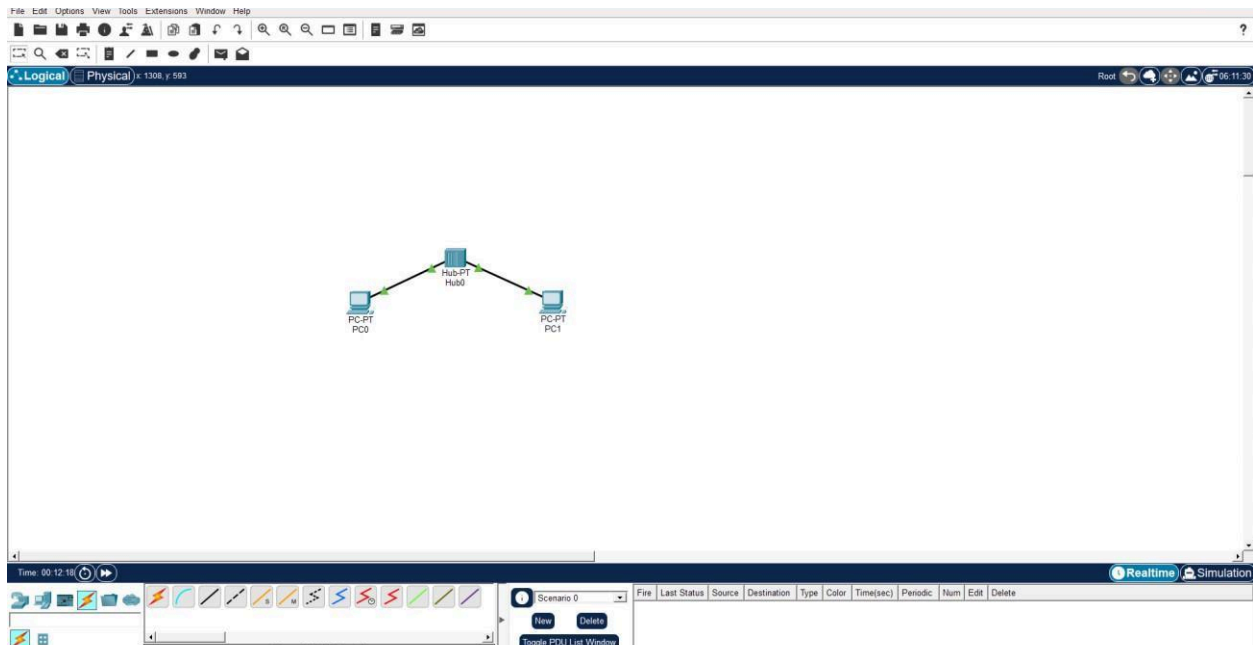
Pinging 10.128.20.1 with 32 bytes of data:

Reply from 10.128.20.1: bytes=32 time<1ms TTL=255
Reply from 10.128.20.1: bytes=32 time<1ms TTL=255
Reply from 10.128.20.1: bytes=32 time<1ms TTL=255
Reply from 10.128.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.128.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

3. Create a network of 2 PCs as below.



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

4. Connect 2 PCs with a Switch below.

The image shows a Cisco Packet Tracer network diagram and a PC command prompt window. The network diagram features a central switch labeled '2960-24TT Switch0' connected to two PCs. The PC on the left is labeled 'PC-PT PC0' with IP address '192.168.1.104'. The PC on the right is labeled 'PC-PT PC1' with IP address '192.168.1.204'. The PC command prompt window is open for PC0, showing a successful ping to 192.168.1.2.

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\> ping 192.168.1.2

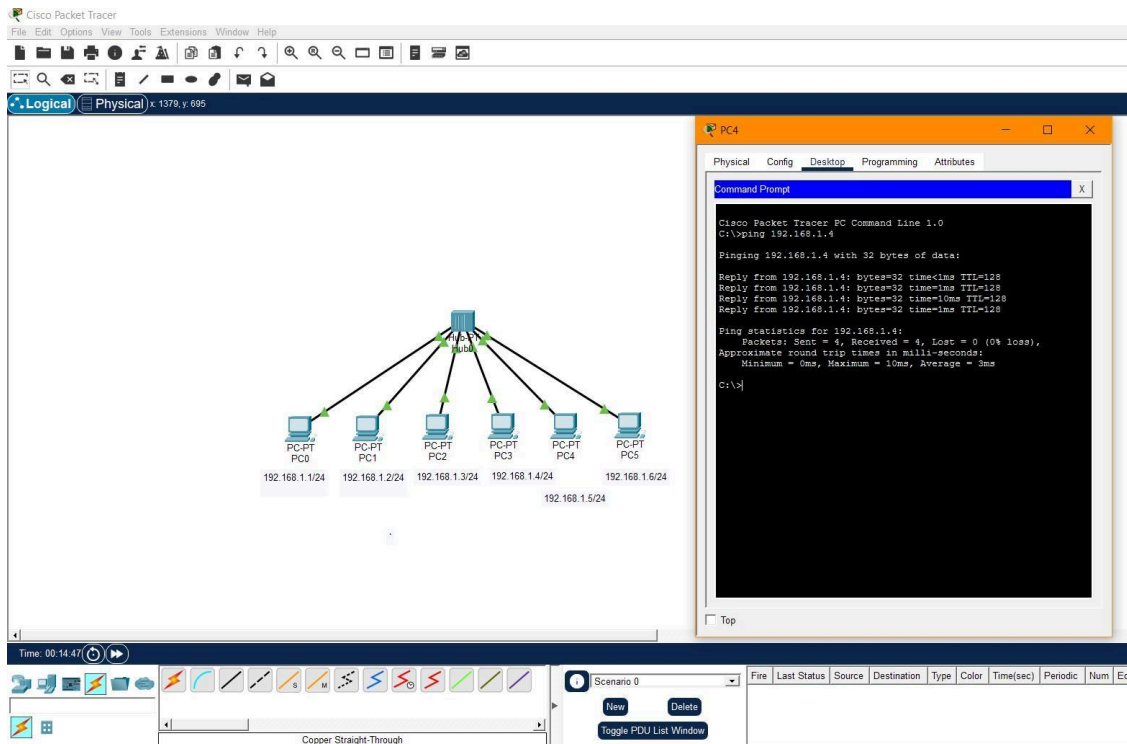
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

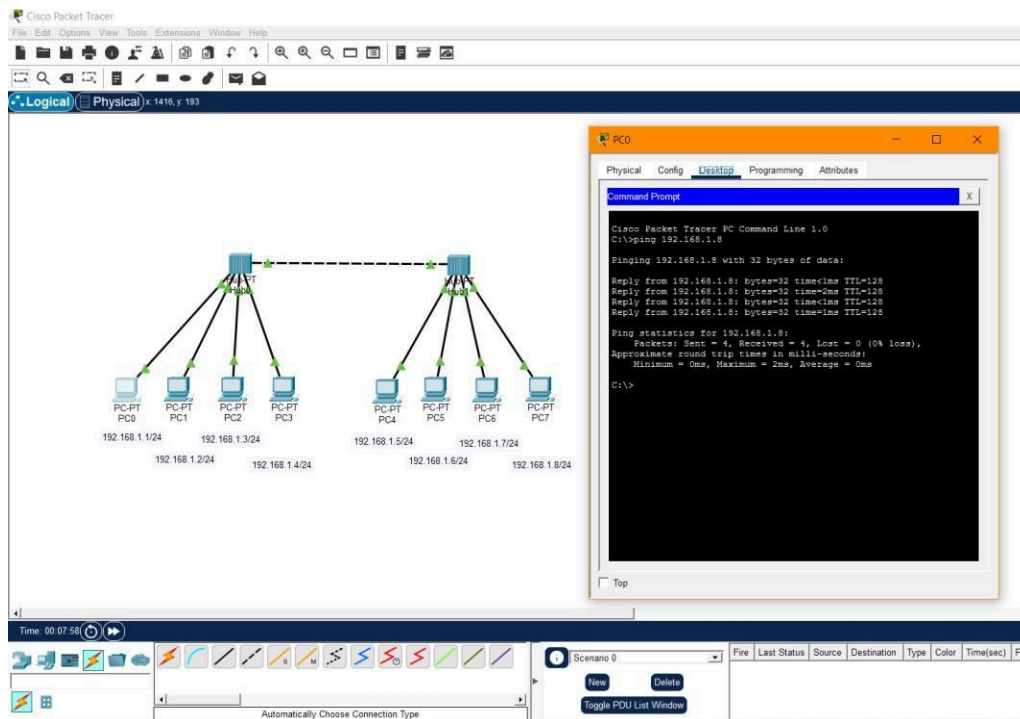
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

5. Create BUS topologies as below.

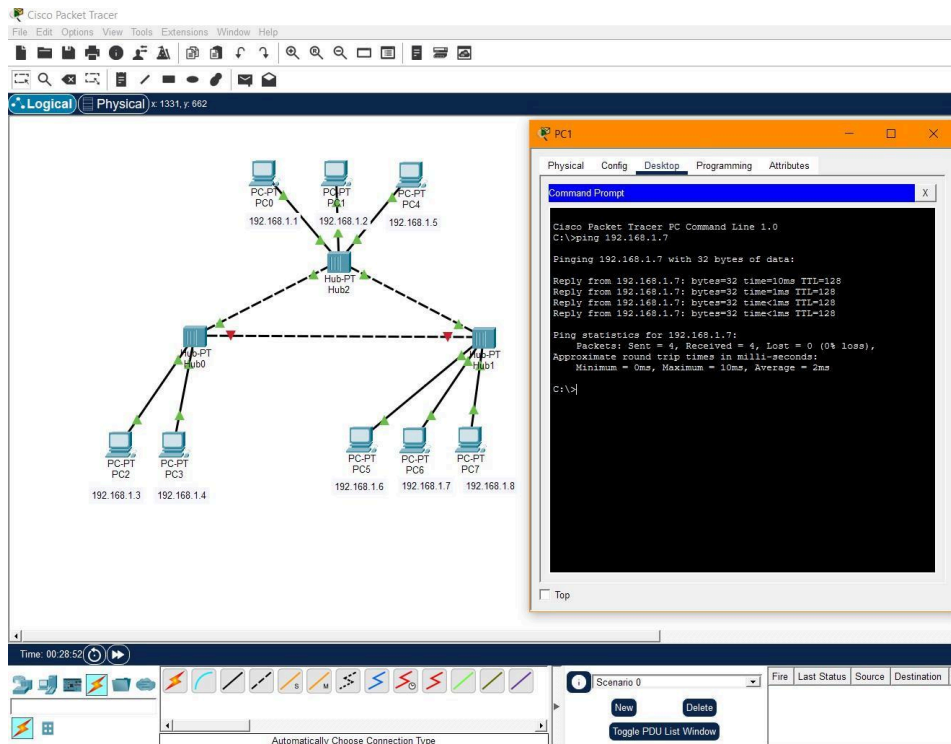


(a)

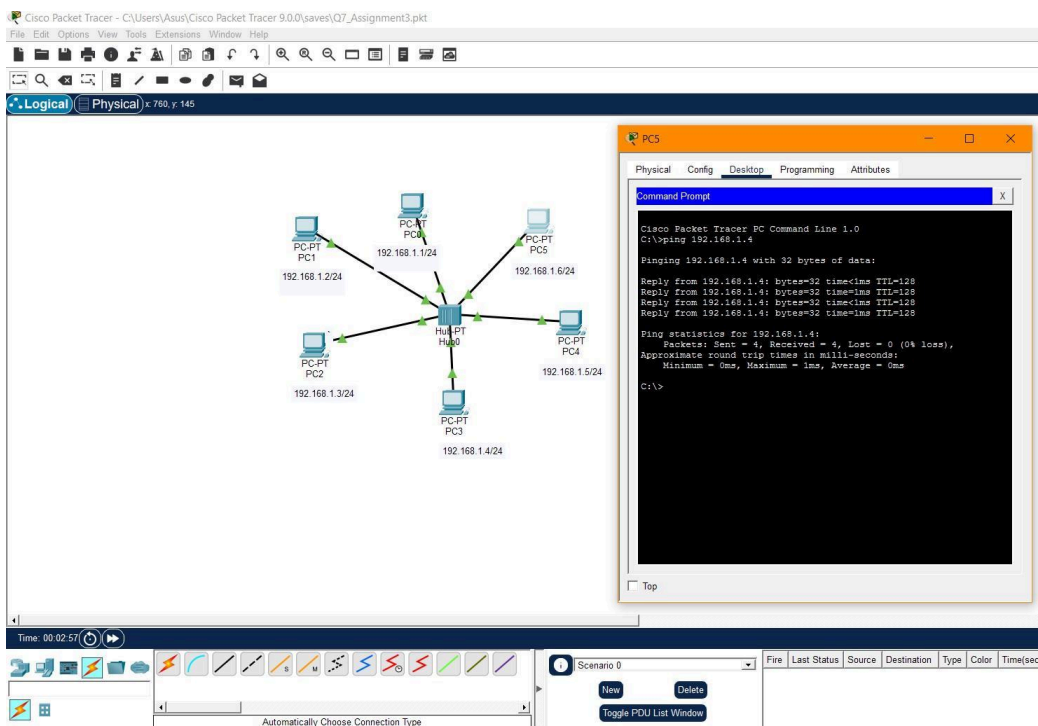


(b)

6. Create the following ring topology



7. Create the following ring topology



8. Create the following ring topology

